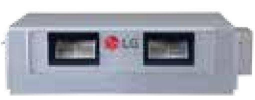
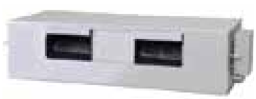
LG

SYSTEM
AIR CONDITIONING

INVERTER DUCTED

INNOVATIVE **AIR CONDITIONING**
SOLUTIONS FOR NEXT-GEN SPACES





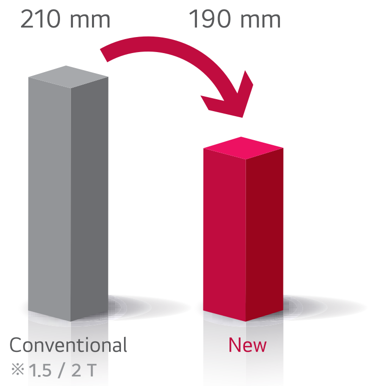
Minimized Height | Easy Maintenance | High Energy Efficiency | Long Distance Piping | Higher Efficiency (BLDC) Motors | Black Fin

Life's Good.

INVERTER LOW STATIC DUCT

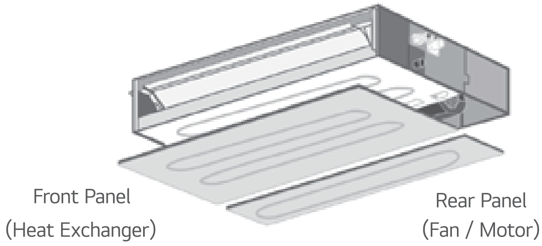
MINIMIZED HEIGHT

New low-static ducts provide ideal solution for installation in limited space.

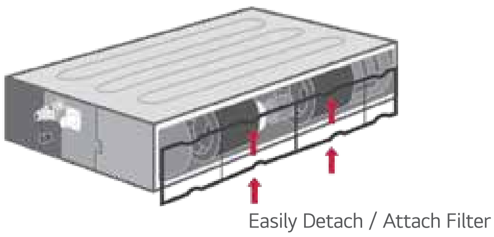


EASY SERVICE & MAINTENANCE

Disassembly of the entire panel is no longer required as the new panel is divided into two parts.
One part is for heat exchanger inspection and the other for fan/motor inspection.
The filter can be easily detached and re-attached even in limited spaces.



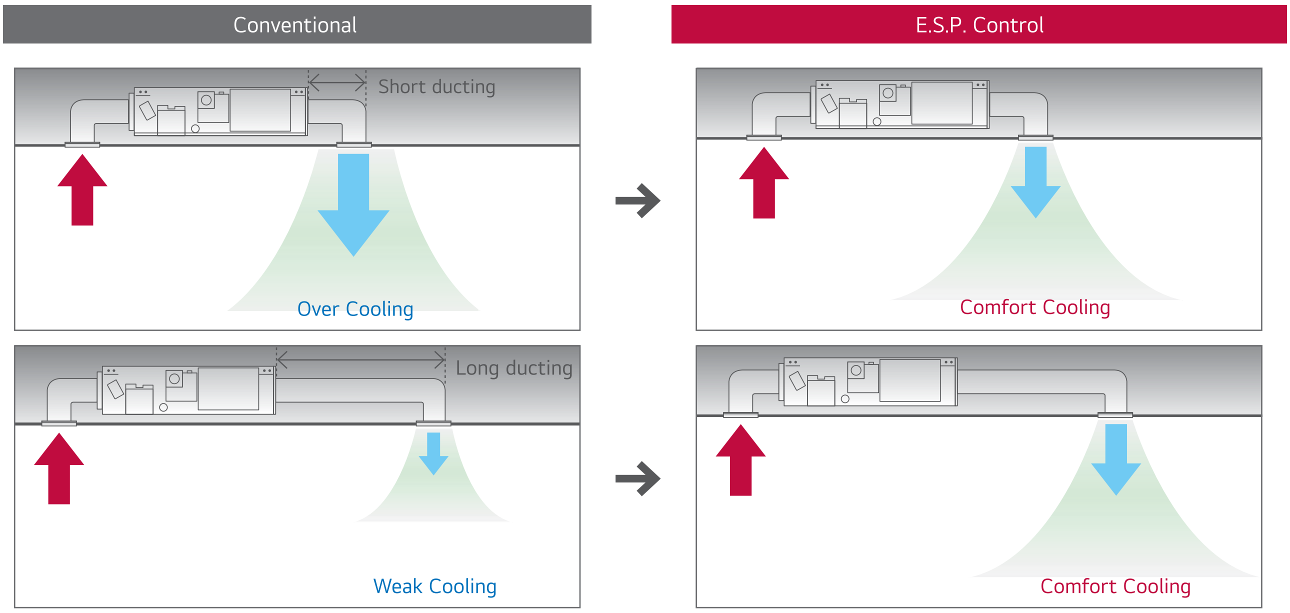
Selective Panel Separation



Easily Detachable Filter

E.S.P. (EXTERNAL STATIC PRESSURE) CONTROL

This function easily controls volume of the air by a remote controller. The BLDC motor can control fan speed and air volume regardless of the external static pressure. Additional accessories are not required to control air flow.

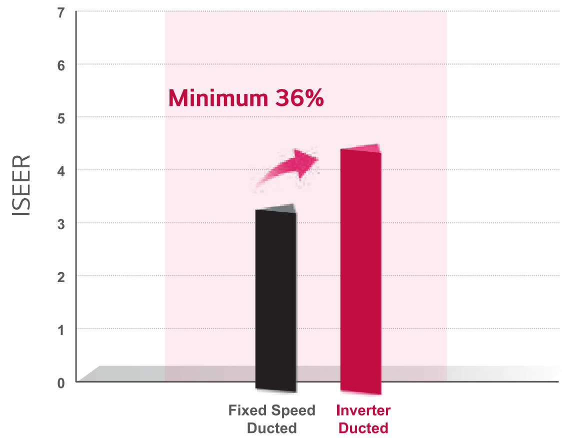


INVERTER HIGH STATIC DUCT

HIGH ENERGY EFFICIENCY

The new Inverter ducted units perform at much higher energy efficiency than conventional ducted splits, since they comprise of Inverter drives, BLDC motors, Electronic expansion valves and other efficiency components and measures explained hereafter.

A comparison of the 2 systems is given below:



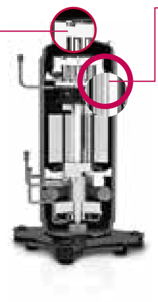
* ISEER value is simulated data as per BEE ISEER Regulation

HIGH ENERGY EFFICIENCY AT PART LOADS

Low load operation efficiency is improved by concentration coil and motor, and 6 By-pass valves

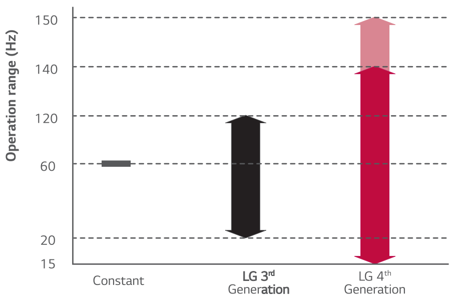
6 By-pass Valves

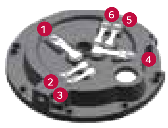
- Compressor reliability is maximised with 6 By-pass valves
- Prevents compressor damage due to excessively compressed refrigerant more efficiently than 4 by-pass valves



Extended Compressor Speed 150Hz

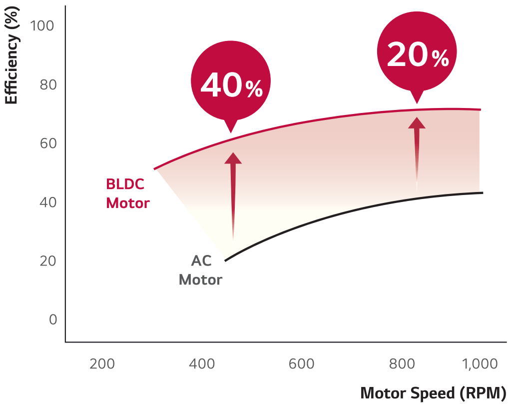
- Rapid operation response
- Capable of reaching required temperature quickly
- Increases part load efficiency





HIGHER EFFICIENCY WITH BRUSHLESS DC (BLDC) MOTORS

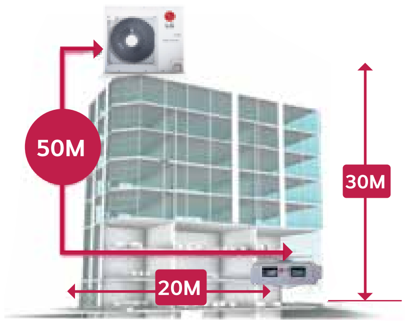
BLDC motors are more efficient than conventional AC motors used by others. BLDC motors are about 40% more efficient at lower speeds and about 20% more efficient at higher speeds.



LONG DISTANCE PIPING

The Inverter ducted units are designed for a 30 m vertical height and a total refrigerant piping distance of 50 m between the indoor and outdoor units.

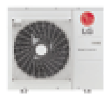
The units are suitable for tall and glass facade buildings also.



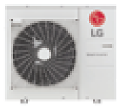
SPECIFICATIONS

INVERTER LOW STATIC DUCT

* Wireless Remote
Controller Included



U18 A Chassis



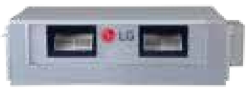
U24 A Chassis

Low Static Duct		Nominal Capacity	T	1.5 T	2.0 T
Combination	Outdoor unit			ZBUQ18GL5A1	ZBUQ24GL6A1
	Indoor unit			ZBNQ18GL5A1	ZBNQ24GL6A1
Rated Capacity	Cooling	Min. ~ Rated ~ Max.	kW	1.58 ~ 5.27 ~ 6.00	2.11 ~ 7.05 ~ 7.74
Power Input	Cooling	Rated	kW	1.64	2.15
EER / COP			W / W	3.21	3.28
Indoor Unit					
Model Name			Unit	ZBNQ18GL5A1	ZBNQ24GL6A1
Power Supply			V , Ø , Hz	230 ,1 , 50	
Air Flow Rate		H / M / L	m³/min	15.0 / 12.5 / 10.0	20.0 / 16.0 / 12.0
Fan Motor	Type			BLDC	
Sound Pressure Level		H / M / L	dB(A)	36 / 34 / 31	39 / 35 / 32
Exterior	Colour		-	Steel Gray	
Dimensions		W x H x D	mm	900 x 190 x 460	1,100 x 190 x 460
Net Weight			kg	19.7	22.0
Shipping Weight			kg	23.6	26.2
Piping Connections	Liquid Side		mm (inch)	Ø 6.35 (1/4)	Ø 9.52 (3/8)
	Gas Side		mm (inch)	Ø 12.7 (1/2)	Ø 15.88 (5/8)
	Drain Pipe	O.D. / I.D.	mm	Ø 32.0 / 25.0	Ø 32.0 / 25.0
Outdoor Unit					
Model Name			Unit	ZBUQ18GL5A1	ZBUQ24GL5A1
Power Supply			V, Ø, Hz	230 ,1 , 50	
Compressor	Type		-	Twin Rotary	
Heat Exchanger	Type (Coating)		-	Fin & Tube Type (Hydrophilic + Black Fin)	
Refrigerant	Type		-	R32	
	Control		-	Electronic Expansion Valve	
Fan Motor	Type			BLDC	
Sound Pressure Level	Cooling	Rated	dB(A)	54	54
Wiring Connections	Power Supply Cable (included Earth)		No. x mm2 (AWG)	3C x 1.5 (14)	3C x 2.5 (12)
Cicuit Breaker			A	16	25
Casing Colour			-	Warm Gray	
Dimensions		W x H x D	mm	770 x 545 x 288	870 x 650 x 330
Net Weight			kg	31.5	41.5 (91.4)
Piping Connections	Liquid	Outer Dia.	mm	Ø 6.35 (1/4)	Ø 9.52 (3/8)
	Gas	Outer Dia.	mm	Ø 12.7 (1/2)	Ø 15.88 (5/8)
Piping Length		Min. / Max.	m	5 (16.4) / 30 (98.4)	5 (16.4) / 50 (164.0)
Maximum Height Difference	Outdoor Unit ~ Indoor Unit	Max.	m	20 (65.6)	30 (98.4)
Operation Range (Outdoor Temperature)	Cooling	Min. ~ Max.	℃ DB	-5 ~ 52	
Accessories*					
Wi-Fi Module				PWFMDD200	
PI485				PMNFP14A1	

Note:
Performances are based on the following condition:
- Cooling: Indoor Ambient Temp. 27° CDB/19°CWB, Outdoor Ambient Temp. 35°CDB/24°CWB
- Interconneced Pipe Length is 7.5m and difference of Elevation (Outdoor ~ Indoor Unit) is 0m.

SPECIFICATIONS

INVERTER HIGH STATIC DUCT



* Wired Remote Controller Included

U4 Chassis

U3 Chassis

U7 Chassis

High Static Duct			Unit	5.5 T	8.5 T	11.0 T
Combination	Outdoor unit			ZBUQ66LRA0	ZBUQ102L8A0	JBUQ132L8A0
	Indoor unit			ZBNQ66LRA0	ZBNQ102L8A0	JBNQ132L8A0
Rated Capacity	Cooling	Min. ~ Rated ~ Max.	kW	5.28 ~17.58 ~ 19.34	8.44 ~ 28.13 ~ 30.94	10.54 ~ 35.16 ~ 38.67
Power Input	Cooling	Rated	kW	6.7	11	10
EER/COP			W / W	2.62	2.56	3.52
ISEER				3.76	3.73	3.89
Indoor Unit						
Model Name			Unit	ZBNQ66LRA0	ZBNQ102L8A0	JBNQ132L8A0
Power Supply			V , Ø , Hz	230, 1, 50		
Air Flow Rate		H / M / L	m3/min	58 / 52 / 46	90 / 78 / 65	113 / 100 / 89
		H / M / L	CFM	2048 / 1836 / 1625	3178 / 2755 / 2295	3991 / 3531 / 3143
Fan Motor	Type		-	BLDC		
Sound Pressure Level		H / M / L	dB(A)	45 / 43 / 41	52 / 50 / 48	49 / 47 / 45
Dimensions	Body	W x H x D	mm	1,230 x 380 x 590	1,562 x 460 x 688	1,562 x 460 x 688
Net Weight	Body		kg	48	81	89
Piping Connections	Liquid		mm	Ø 9.52 (3/8)	Ø 9.52 (3/8)	Ø 12.7 (1/2)
	Gas		mm	Ø 19.05 (3/4)	Ø 22.2 (7/8)	Ø 22.2 (7/8)
	Drain (O.D. / I.D.)		mm	Ø 25.4 / 19.4	Ø 25.4 / 21.0	Ø 25.4 / 21.0
Power and Communication Cable (included Earth)		No. x mm2 (AWG)		4C x 1.0 (18)	4C x 1.0 (18)	4C x 1.0 (18)
Outdoor Unit						
Model Name			Unit	ZBUQ66LRA0	ZBUQ102L8A0	JBUQ132L8A0
Power Supply			V , Ø , Hz	415, 3, 50		
Compressor	Type	-	-	Twin Rotary	Twin Rotary	Scroll
Heat Exchanger	Type (Coating)			Fin & Tube Type (Hydrophilic + Black Fin)		
Refrigerant	Type		-	R32	R32	R410A
	Control		-	Electronic Expansion Valve		
Fan Motor	Type			BLDC		
	Output		W x No.	124 x 1	124 x 2	124 x 2
Sound Pressure Level		Rated	dB(A)	55	66	63
Wiring Connections	Power Supply Cable (included Earth)		No. x mm2 (AWG)	5C x 4.0 (12)	5C x 6.0 (10)	5C x 6.0 (8)
Circuit Breaker			A	20	40	40
Casing Colour			-	Warm Gray		
Dimensions		W x H x D	mm	950 × 834 × 330	950 × 1,380 × 330	1,090 ×1,625 × 380
Net Weight			kg	76	100	150
Piping Connections	Liquid	Outer Dia.	mm	Ø 9.52 (3/8)	Ø 9.52 (3/8)	Ø 12.7 (1/2)
	Gas	Outer Dia.	mm	Ø 19.05 (3/4)	Ø 22.2 (7/8)	Ø 22.2 (7/8)
Piping Length		Max.	m	50 (164.0)	50 (164.0)	50 (164.0)
Maximum Height Difference	Outdoor Unit ~ Indoor Unit	Max.	m	30 (98.4)	30 (98.4)	30 (98.4)
Operation Range (Outdoor Temperature)	Cooling	Min. ~ Max.	℃ DB	-5 ~ 53	-5 ~ 53	-5 ~ 53
Accessories*						
Wi-Fi Module				PWFMDD200		
PI485				PMNFP14A1		

Note:
Performances are based on the following condition:
- Cooling: Indoor Ambient Temp. 27° CDB/19°CWB, Outdoor Ambient Temp. 35°CDB/24°CWB
- Interconneced Pipe Length is 7.5m and difference of Elevation (Outdoor ~ Indoor Unit) is 0m.
- R32 available in all models except 11T Duct

BLACK FIN FOR CORROSION RESISTANCE

ANTI CORROSION COATING

BLACK FIN

Black fin for corrosion resistance

Hydrophilic coating
The Hydrophilic Coating minimises moisture build up on the fin.

Complex Resin (Corrosion resistant)
The black coating provides strong protection from corrosion.

Aluminum Fin

CORROSION RESISTANCE PROVEN BY CERTIFIED TESTS

LG Corrosion Resistance solution passed ISO accelerated corrosion test conducted by an independent test organization and the result has been certified by prestigious global certification organization, TUV.

※ Verification of corrosion resistance performance
- Test Method B of ISO21207
- ASTM B117 / ISO 9227 (10,000 hours)

LINE UP

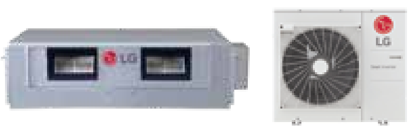
INVERTER LOW STATIC DUCT

R32

1.5 T	2.0 T
	

INVERTER HIGH STATIC DUCT

R32

5.5 T	8.5 T	11.0 T*
		

*R32 available in all models except 11T Duct